ArohaSetu

Notice Sooner, Care Better.

Healthcare, Wellbeing and Elderly Care

Problem Statement / Concept Title: Privacy-Conscious Patient Monitoring and Emergency Response

Project: ArohaSetu
Reg. No.: HSC|AN|00004

From Delayed Awareness to Structured Escalation

Status Key: **IMPLEMENTED** in prototype · **SIMULATED** in test harness · **PLANNED** for production hardware



PROBLEM 01

Unobserved Fall or Decline

Older adults living alone risk acute injury or deterioration without immediate help.

IMPLEMENTED

Camera Pose Analysis: OpenCV + Ultralytics YOLO-Pose detects fall and immobility states locally.



PROBLEM 02

Remote Family Uncertainty

Families living far away lack real-time visibility into an elder's immediate condition.

IMPLEMENTED

Caregiver Notification: Dedicated web dashboard with configured alerts and active status feeds.



PROBLEM 03

Ward Monitoring Bottlenecks

Clinical teams struggle to watch multiple dispersed patients continuously.

IMPLEMENTED

Virtual-Ward Triage: Nurse workflow engine running deterministic NEWS2 clinical risk scores.



PROBLEM 04

Geographic Care Latency

Island and rural communities encounter transport and communication delays reaching care.

SIMULATED / PLANNED

Dual-Mode Triage: Home mode paired with hardware sensor integration (Planned) and offline app (Implemented).



Core Innovation: Combines physiological validation with privacy-conscious on-device pose estimation and deterministic clinical scoring under home and virtual-ward modes. *No claim of absolute prevention or guaranteed earlier diagnosis.*

Modular End-to-End Pipeline & Implementation Stack

01

01 · Data Inputs

Sensor Vitals: Heart rate, SpO2, blood pressure (**SIMULATED** in demo; hardware sensors **PLANNED**).

Vision Stream: Local camera feed for frame-by-frame pose analysis (**IMPLEMENTED**).

02

02 · Processing

Vision Engine: Python, OpenCV, Ultralytics YOLO-Pose extracting skeletal keypoints locally without raw video export.

Signal Validation: Boundary checking and noise filtering on ingested vitals.

03

03 · Risk Evaluation

NEWS2 Engine: Deterministic scoring against Royal College of Physicians clinical thresholds.

Fall State Engine: Sudden elevation drop and post-fall immobility verification windows.

04

04 · Interfaces

Web Dashboard: React + Tailwind CSS with Node.js/Express REST backend.

Mobile Client: Kotlin + Jetpack Compose app supporting local offline alert caching.

Repository & Live Demo:

GitHub: github.com/EternalFlames131/Arohasetu

Live Web Demo: arohasetu.vercel.app

AI Usage Disclosure:

Gamma was used for slide layout. AI assistance was limited to light coding help and bug checks; the project concept and implementation are team-created.

Ecosystem Connectivity & Intended Benefits

Intended benefits, not proven outcomes



People & care teams

Older adults living alone; family caregivers; nurses and hospital teams; primary healthcare workers; isolated communities.



Intended benefits

Earlier awareness of concerning vital-sign changes; faster caregiver notification on configured alerts; better visibility of monitoring and follow-up; less uncertainty for remote families; potential reach to underserved communities.



Stakeholders

Patients and older adults; families and caregivers; nurses and healthcare institutions; community health workers; emergency-response services; public healthcare administrators.

Connected ecosystem: Patients & older adults ➔ Families & caregivers ➔ Nurses & healthcare institutions ➔ Community health workers ➔ Emergency-response services ➔ Public healthcare administrators

Prototyped Feasibility & Six Managed Risk Factors

Software components validated iteratively; risks are transparently acknowledged and active, not claimed as resolved

01 Clinical Accuracy

Risk: Algorithmic misinterpretation of physiological instability.
Mitigation: Use standardized NEWS2 clinical guidance; mandate clinician-in-the-loop oversight.

02 Island Network Connectivity

Risk: Intermittent remote Internet connection in remote island geography.
Mitigation: Local edge alert caching on Kotlin Android client; explicit indicators when cloud sync is offline.

03 False Alarms

Risk: Alert fatigue caused by transient movement artifacts.
Mitigation: Temporal verification windows; rule hysteresis; multi-frame keypoint persistence checks.

04 Emergency Escalation Failures

Risk: External ambulance dispatch services failing to ingest automated alerts.
Mitigation: Explicit simulation labeling; strict protocol verification before live agency integration.

05 Patient Privacy

Risk: Reluctance to place optical sensors in domestic private spaces.
Mitigation: Strictly on-device keypoint inference; no raw video retention or transmission; access controls.

06 Clinical & Regulatory Readiness

Risk: Premature deployment without biomedical software certification.
Mitigation: Dedicated clinical governance, ethical review, and compliance validation prior to clinical trials.

Feasibility Status: Web simulator, local Python pose detector, and offline-capable Android APK functional for demonstration; live physical telemetry requires field development.

Team Composition & Development Journey

MEMBERS & ROLES

Team Structure

Team Composition:
Samrat Dasgupta - Team Leader / Project Idea
Hriday Das - Main Developer / Coder
C. Mohd. Jasheem - Designer

PROJECT PURPOSE

ArohaSetu Care Model

A personal nurse for every family.

Remote monitoring supports care at home and in hospital virtual wards. Wearable vitals and privacy-first camera zones help surface concerns to caregivers.

Portal Registration: Project: ArohaSetu · Registration: HSC|AN|00004 · Andaman & Nicobar Islands

Clinical Foundations & Project Artefacts

CLINICAL & COMPETITION STANDARDS

1. Clinical Standard

Royal College of Physicians. *National Early Warning Score (NEWS) 2: Standardising the assessment of acute-illness severity in the NHS.* London: RCP, 2017.
rcp.ac.uk/resources/national-early-warning-score-news-2

2. Competition Framework

Hack for Social Cause: *Hackathon Presentation Template v7.0*, official submission instructions.

CODEBASE & EVIDENCE INTEGRITY

3. Project Repositories

Source Code: github.com/EternalFlames131/Arohasetu
Interactive Web Simulator: arohasetu.vercel.app

4. Data & Claims Integrity

No external statistics are used.